SOMAIYA VIDYAVIHAR UNIVERSITY
K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

Somaiya TRUST

Module 5 - Uncertain Knowledge and Reasoning

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Outline

- Uncertainty
- Bayesian Networks
  - Syntax
  - Global Semantics
  - Local Semantics
- Constructing Bayesian Networks

Introduction to Uncertainty

- Why uncertainty exists in Intelligent System?
- Example – Taxi Driver
- **Key Idea**
  - Artificial Intelligence systems must **reason with incomplete or uncertain information**.
- **Uncertainty** refers to situations where the agent does not have complete knowledge about the environment.

Characteristics of knowledge

- Voluminous
- Hard to characterise
- Constantly changing

- Uncertainty means that many of the simplifications that are possible with deductive inference are no longer valid.

Example- taxi driver

Performance measure

- -2 per breaking signal
- +10 reaching in time
- -1 per missed speed breaker
- +8 choosing optimal route
- -6 reaching too early

- Plan 1- leaving 100 minutes early for a distance of 10 kms and driving at reasonable speed
- Plan 2 – leaving 300 minutes early
- Plan 3- leaving 60 mins early and drive like crazy

Uncertainty

- Car breaks down
- Runs out of fuel,
- Agent gets into an accident,
- There are accidents on the bridge,
- Someone had wrong inventory and now the ordered goods are not in warehouse
- Strike/disruption on the way to warehouse/delivery
- .....

- None of these conditions can be deduced, so the plan's success cannot be inferred.

Example – Dental Diagnosis

∀p Symptom(p, Toothache) ➔ Disease(p, Cavity) .

➔ Wrong

- Not all patients with toothaches have cavities

∀ p Symptom(p, Toothache) ➔ Disease(p, Cavity) V Disease(p, GumDisease) C' Disease(p, Abscess) . . .

- Almost unlimited list of possible causes

∀ p Disease(p, Cavity) ➔ Symptom(p, toothache) ➔ Wrong

Not all cavities cause pain. Only logically exhaustive list may help.

- Still there's possibility that a patient might have a toothache and a cavity that are unconnected.

Reasons which summarize uncertainty

- **Laziness:** It is too much work to list the complete set of antecedents or consequents needed to ensure an exceptionalness rule and too hard to use such rules.
- **Theoretical ignorance:** There may not any complete theory for the domain. E.g. Medical Science
- **Practical ignorance:** Even if we know all the rules, we might be uncertain about the problem instance in hand.
- These reasons are inescapable in complex, dynamic, or inaccessible worlds.

Uncertainty … Contd

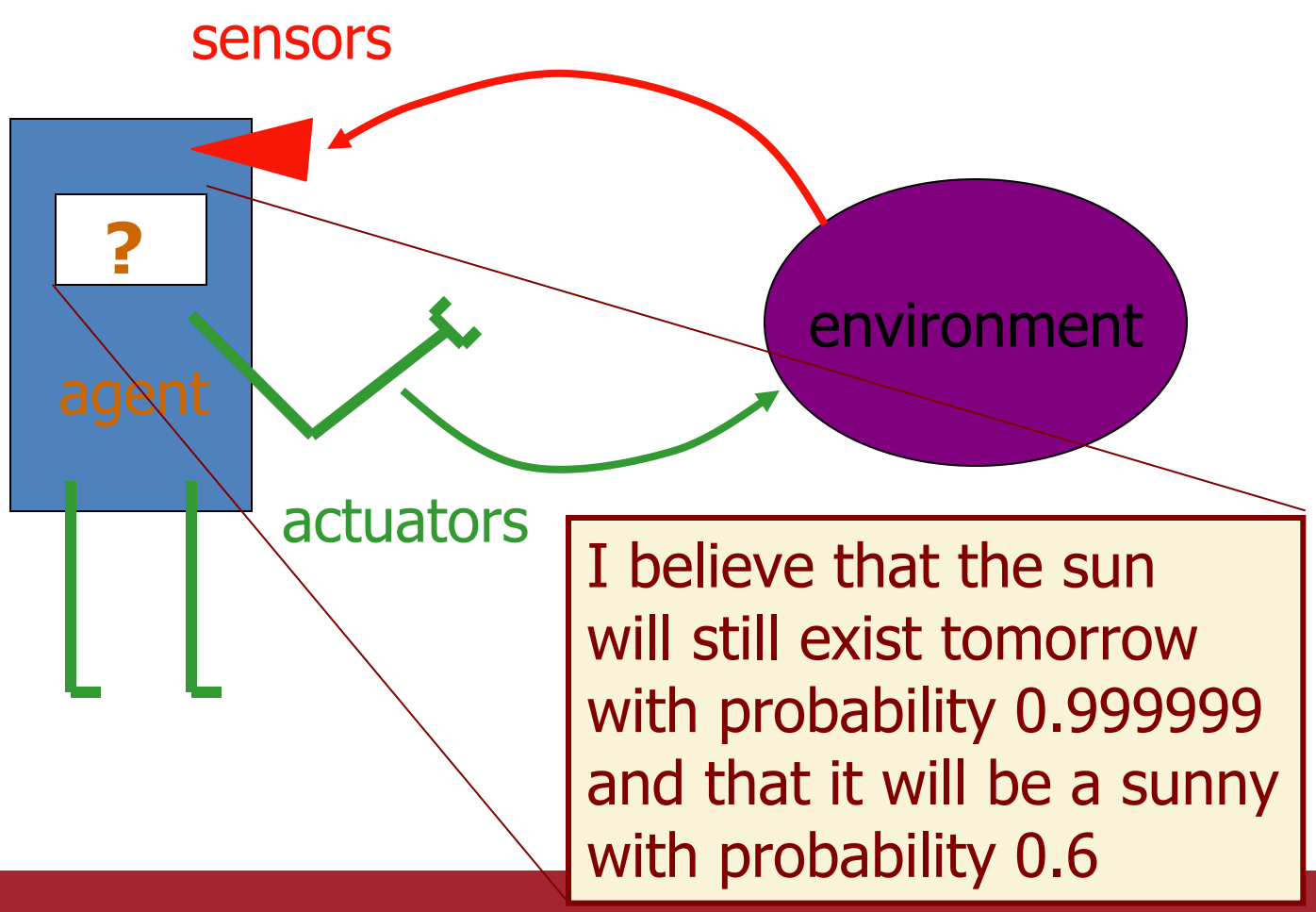
- Same theories as medical science may be applicable to law, business, design, automobile repair, gardening, dating, and so on.
- The agent's knowledge can at best provide only a **degree of belief** in the relevant sentences.
- Solution - Main tool for dealing with degrees of belief will be **probability theory,** which assigns to each sentence a numerical degree of belief between 0 and 1.
- *Probability provides a way of* **summarizing** *the uncertainty that comes from our laziness and ignorance.*
- *E.g.* patient has a cavity if he or she has a toothache.

Decision Theory

- Utility theory says that every state has a degree of usefulness, or utility, to an agent and that the agent will prefer states with higher utility.

Decision theory = probability theory + utility theory .

Probabilistic Agent



Problem

- At a certain time t, the KB of an agent is some collection of beliefs
- At time t the agent's sensors make an observation that changes the strength of one of its beliefs
- How should the agent update the strength of its other beliefs?

Purpose of Bayesian Networks

- **Bayesian network** represents the dependencies among variables and to give a concise specification of any full joint probability distribution.
- Facilitate the description of a collection of beliefs by making explicit causality relations and conditional independence among beliefs
- Provide a more efficient way (than by using joint distribution tables) to update belief strengths when new evidence is observed

Other Names

- Belief networks
- Probabilistic networks
- Causal networks

Who's Bayes? 1702-1761

- **English theologian** and mathematician Thomas Bayes has greatly contributed to the field of probability and statistics. **His ideas have created much controversy and debate among statisticians over the years**….
- Bayes wrote a number of papers that discussed his work. However, **the only ones known to have been published while he was still living are:** ***Divine Providence and Government Is the Happiness of His Creatures*** **(1731) and** ***An Introduction to the Doctrine of Fluxions, and a Defense of the Analyst*** (1736)…

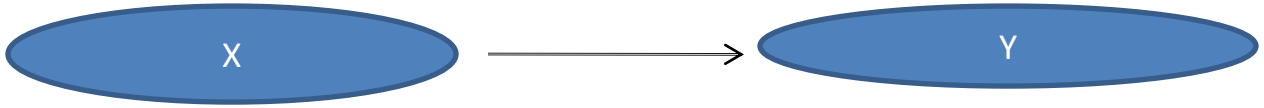
Thomas Bayes

http://mnstats.morris.umn.edu/introstat/history/w98/Bayes.html

Specification of Bayesian/Belief Network

A belief network is a graph with the following:

1. Nodes : A set of random variables.
2. Directed links : directed links or arrows connects pair of nodes.

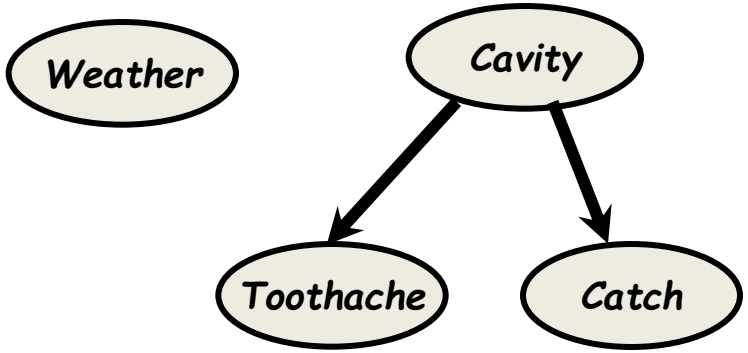


X is the Parent of Y.

3. Each node X_i has a **conditional probability** distribution $P(X_i|Parent(X_i))$ that quantifies the effect of the parents on the node.
4. The graph has no directed cycles (DAG).

Example 1

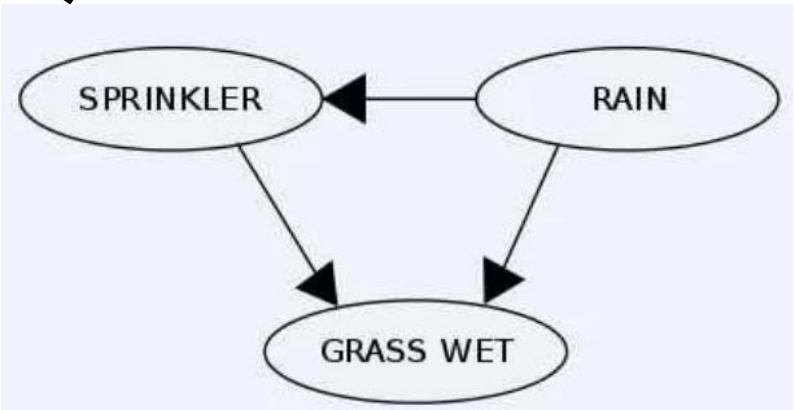
Topology of network encodes conditional independence assertions:



Weather is independent of other variables
Toothache and Catch are conditionally independent given Cavity

Example 2 - BBN

- Consider a simple BBN with three variables: **Rain (R)**, **Sprinkler (S)**, and **Wet Grass (W)**.

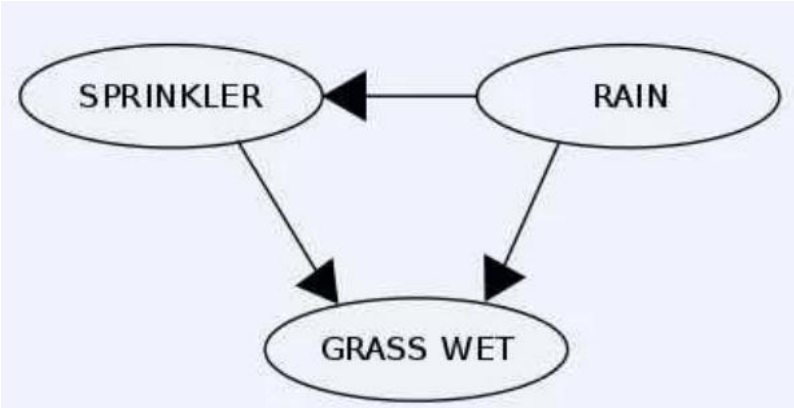


- The network encodes the following relationships:
  - The sprinkler depends on whether it rains: P(S|R)
  - The grass being wet depends on both the sprinkler and rain: (PW|R,S)

Example 2 – BBN (continued)

- Each node in the network contains a **Conditional Probability Table (CPT)**, which defines the probability distribution of the variable given the states of its parent nodes.

Rain	Sprinkler
F	0.4
T	0.01



Rain
0.2

Sprinkler	Rain	Wet
T	T	0.99
T	F	0.9
F	T	0.8
F	F	0.01

The Joint Probability Distribution

- The joint probability of the entire network is expressed as the product of the **local conditional probabilities** of each variable, given its parent variables. This decomposition reduces the computational burden significantly.
- A generic entry in the joint probability is the probability of a conjuction of a particular assignments to each variables. distribution $P(x_1,\ldots,x_n)$ is given by:

$$\boldsymbol{P}\,(X_1, \ldots, X_n) = \pi_{i=1}\, \boldsymbol{P}\,(X_i \mid Parents(X_i))$$

Example 2 – BBN (contd...)

- What is the probability that it is raining, given the grass is wet?

  P(W,S,R)

- Joint Probability

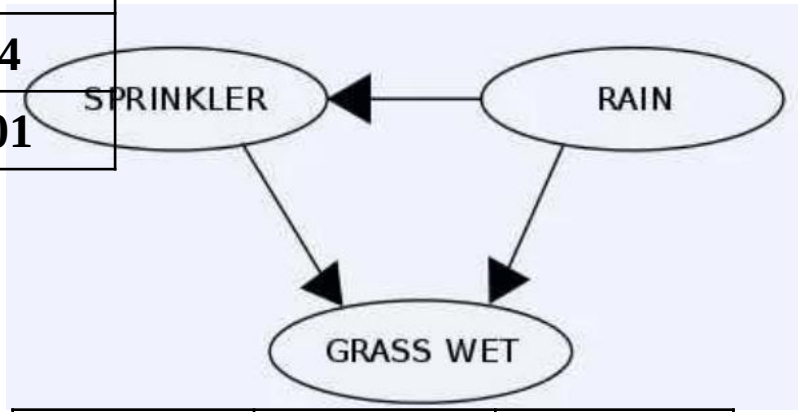
$$P(W,S,R)=P(W|S,R)\cdot P(S|R)\cdot P(R)$$

$$=0.99 \text{ X } 0.01 \text{ X } 0.2$$

$$=0.00198$$

Rain
0.2

Rain	Sprinkler
F	0.4
T	0.01



Sprinkler	Rain	Wet
T	T	0.99
T	F	0.9
F	T	0.8
F	F	0.01

Example 3 BBN

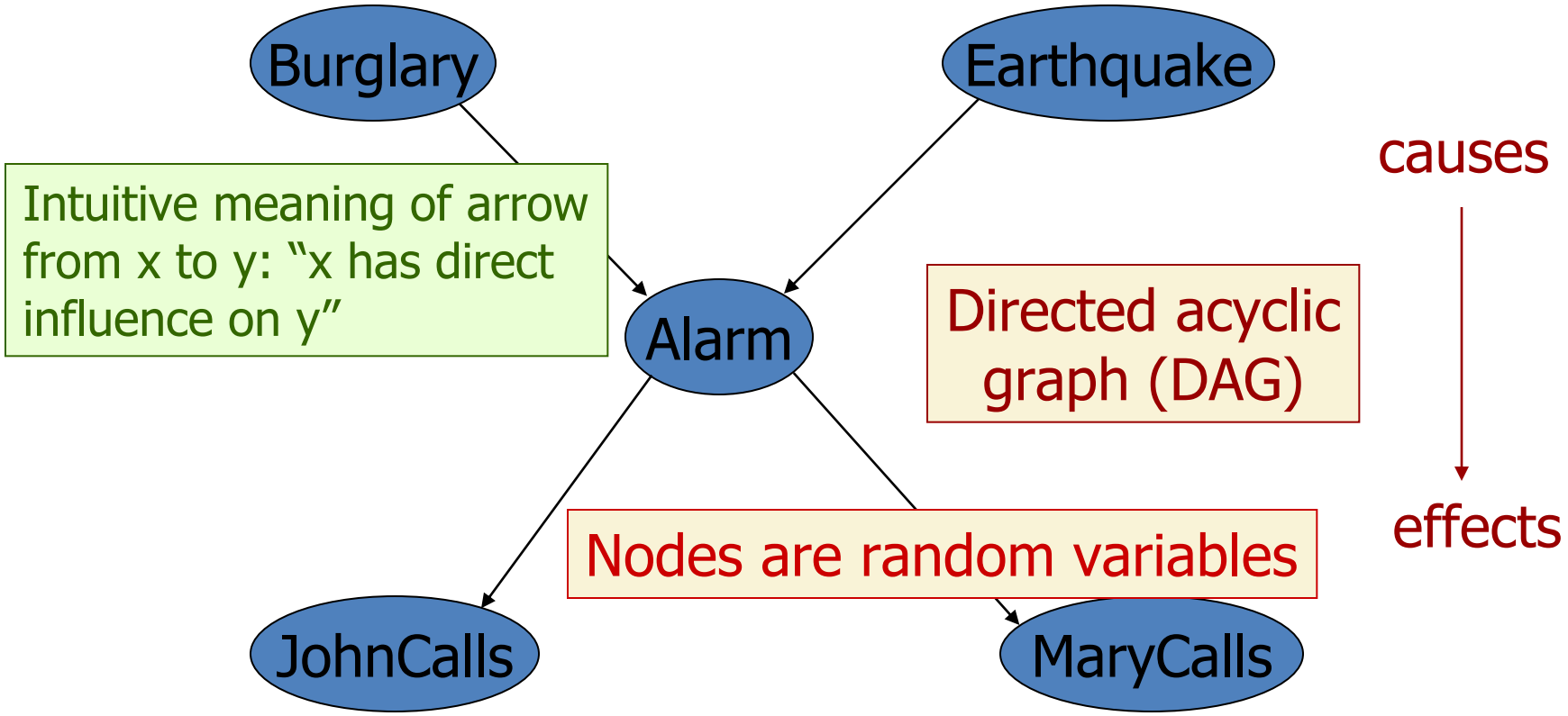
- You have a new burglar alarm installed at home. It is fairly reliable at detecting a burglary, but also responds on occasion to minor earthquakes. You also have two neighbours, John and Mary, who have promised to call you at work when they hear the alarm. John always calls when he hears the alarm, but sometimes confuses the telephone ringing with the alarm and calls then, too. Mary, on the other hand, likes rather loud music and sometimes misses the alarm altogether.

Example 3 (contd..)

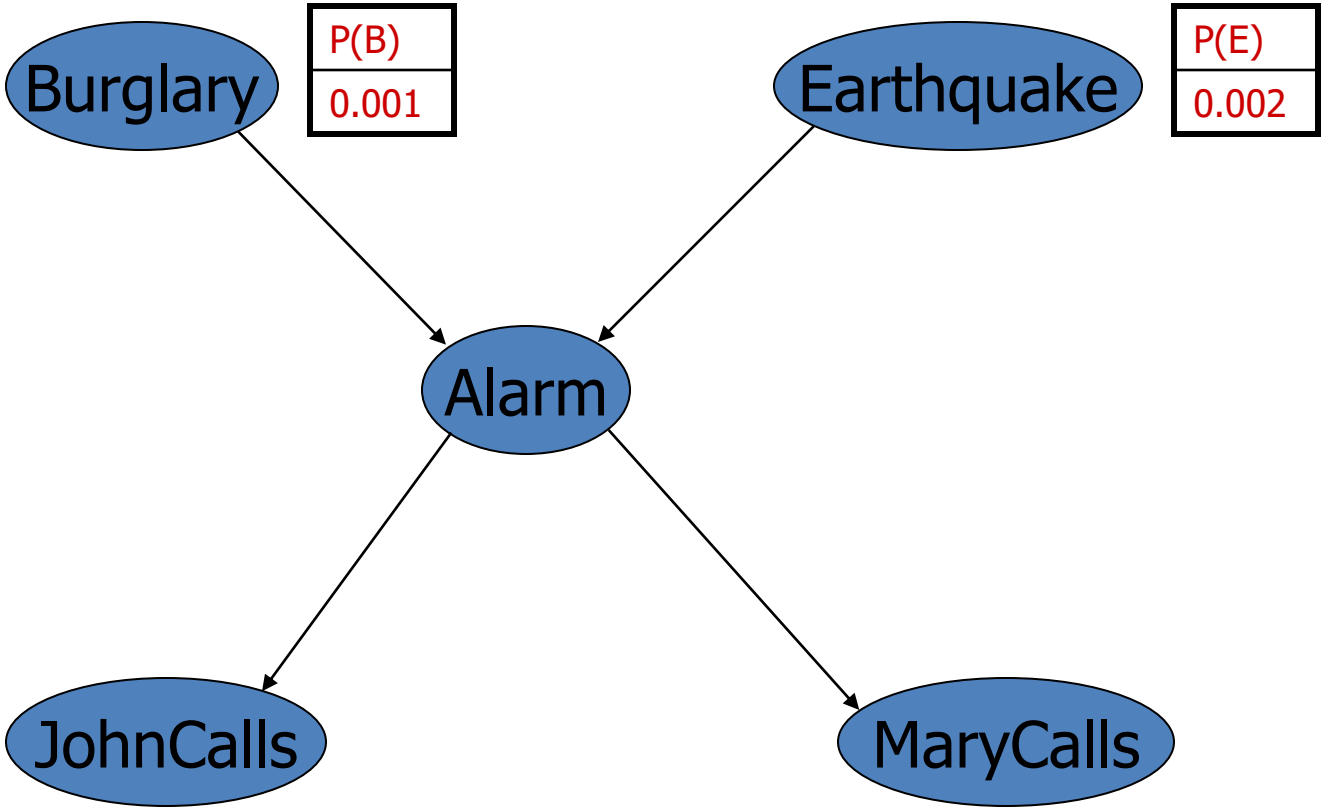
Boolean Variables: Burglary, Earthquake, Alarm, JohnCalls, MaryCalls

Network topology reflects "causal" knowledge:
- A burglary can set the alarm off
- An earthquake can set the alarm off
- The alarm can cause Mary to call
- The alarm can cause John to call

A Simple Belief Network



Assigning Probabilities to Roots



Conditional Probability Tables

CPTs:

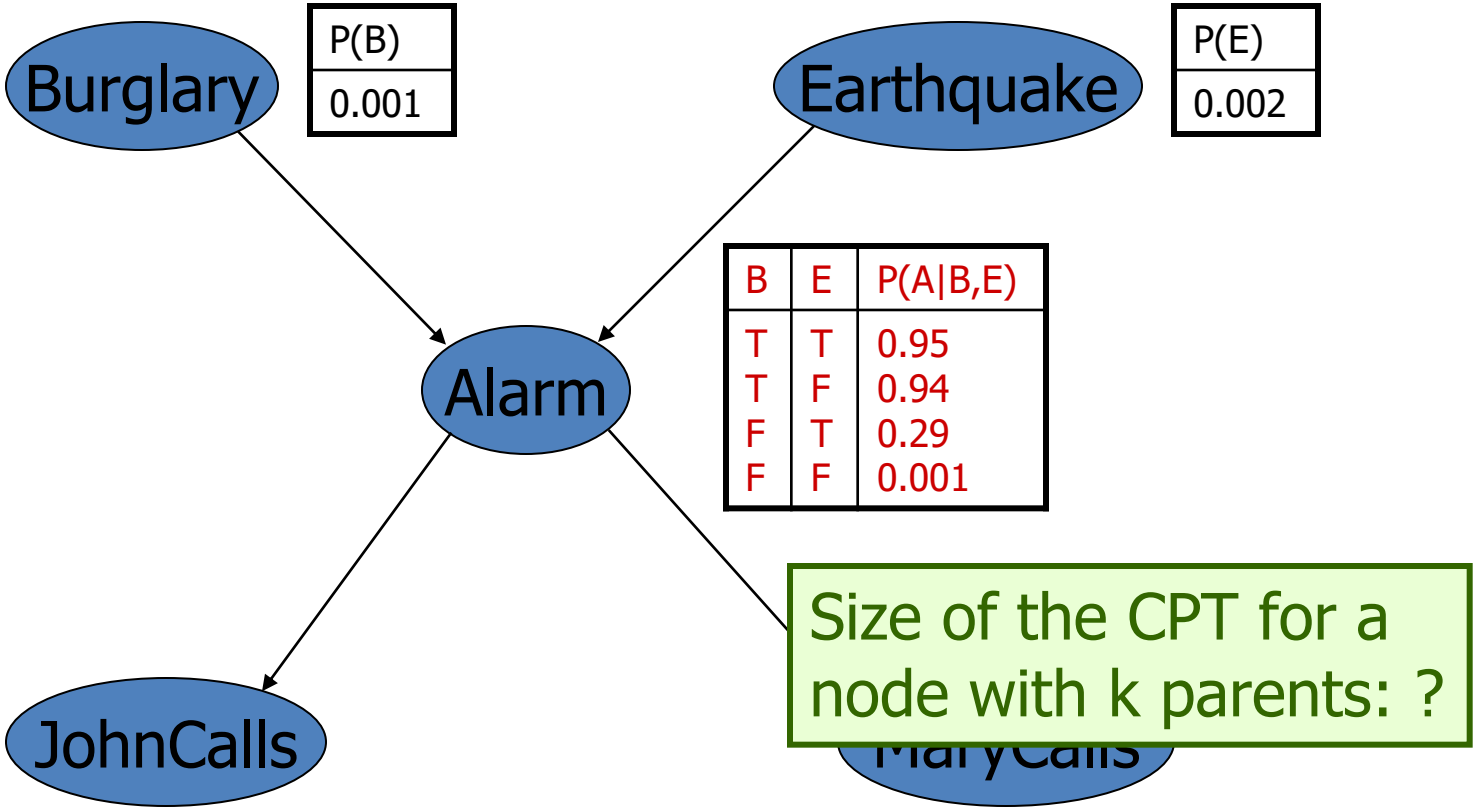
- alarm setoff if burglar in 94% of cases
- alarm setoff if earthquake in 29% of cases
- false alarm setoff in 0.1% of cases
- Notice: in CPTs like P(A|B), only P(a|B) are reported, because P(¬a|B) = 1 – P(a|B)

Conditional Probability Tables (contd..)

- Each node X_i in a belief network has a CPT expressing the probability of X_i, given its parents as evidence.

CPT for Alarm	Burglary	Earthquake	Alarm	
			T	F
	T	T	0.950	0.050
	T	F	0.940	0.060
	F	T	0.290	0.710
	F	F	0.001	0.999

Conditional Probability Tables



P(B)
0.001

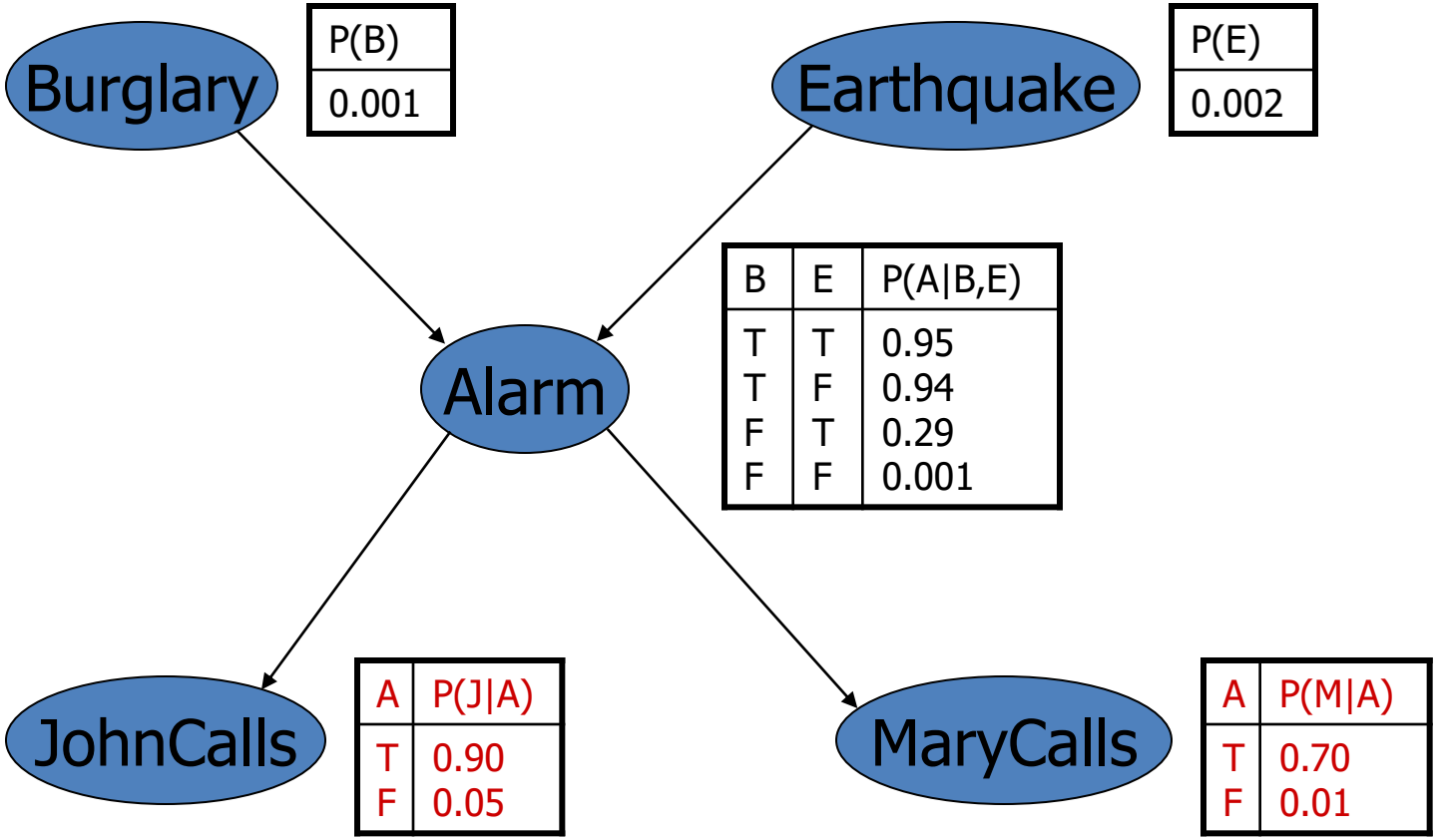
P(E)
0.002

B	E	P(A|B,E)
T	T	0.95
T	F	0.94
F	T	0.29
F	F	0.001

Size of the CPT for a node with k parents: ?

Note: the tables only show P(X) here because P(¬X) = 1 – P(X)

Conditional Probability Tables



P(B)
0.001

P(E)
0.002

B	E	P(A|B,E)
T	T	0.95
T	F	0.94
F	T	0.29
F	F	0.001

A	P(J|A)
T	0.90
F	0.05

A	P(M|A)
T	0.70
F	0.01

Semantics of Bayesian network

There are two, equivalent, ways to interpret a belief network for the variables $X_1, \ldots, X_n$:

1. To see the network as a representation of the joint probability distribution $P(X_1, \ldots, X_n)$.
2. View as an encoding of a collection of conditional independence statements about $X_1, \ldots, X_n$.

Interpretation 1 is helpful in understanding how to *construct belief networks*.

Interpretation 2 is helpful in *designing inference procedures* on top of them.

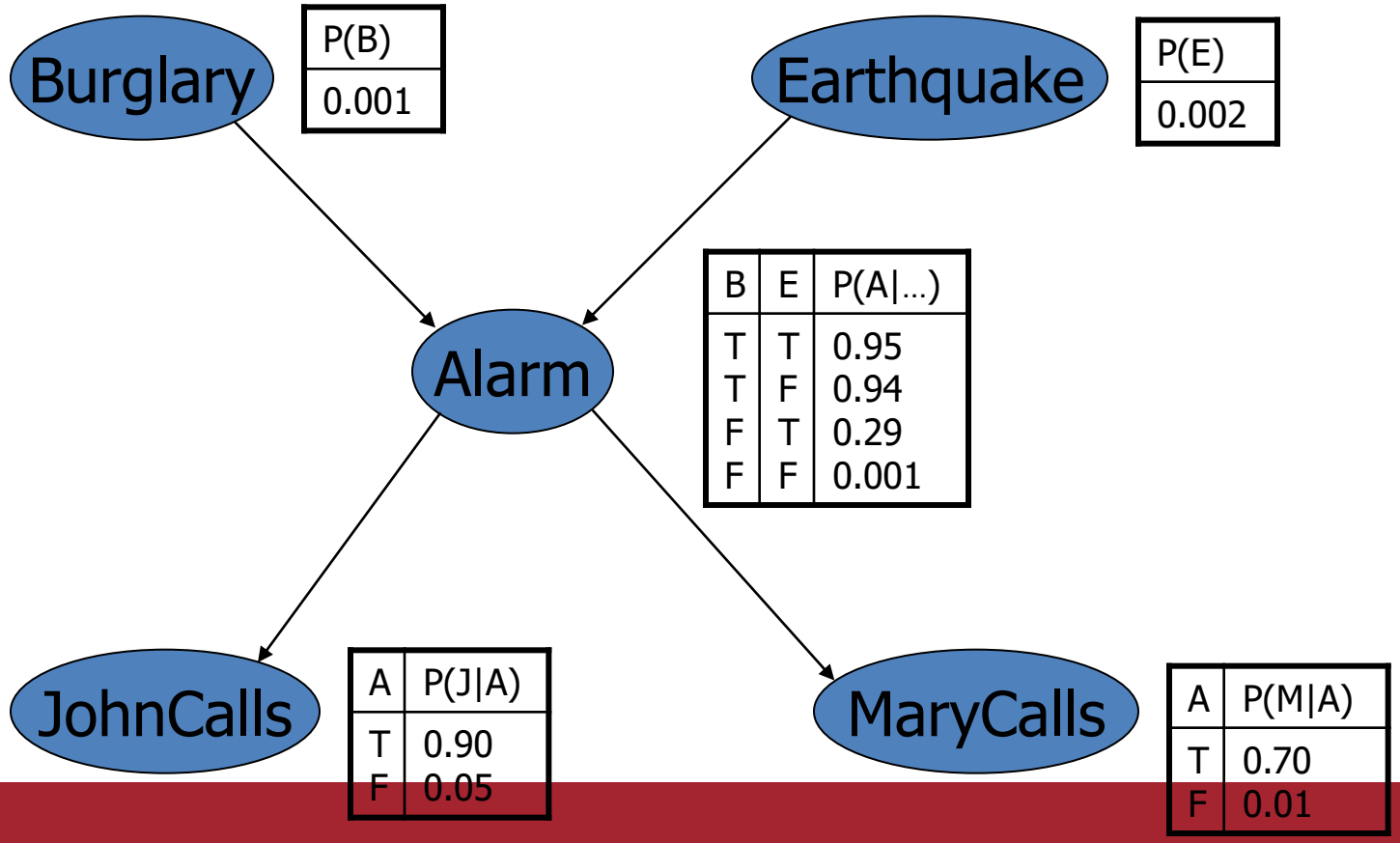
The Joint Probability Distribution

- A generic entry in the joint probability is the probability of a conjuction of a particular assignments to each variables. distribution $P(x_1,\ldots,x_n)$ is given by:

$$\boldsymbol{P}(X_1, \ldots, X_n) = \Pi_{i=1} \boldsymbol{P}(X_i \mid Parents(X_i))$$

BN as Joints

$$P(x_1,x_2,\ldots,x_n) = \Pi_{i=1,\ldots,n} P(x_i|Parents(X_i))$$



P(B)
0.001

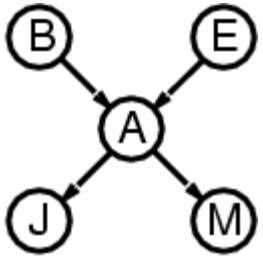
P(E)
0.002

B	E	P(A|...)
T	T	0.95
T	F	0.94
F	T	0.29
F	F	0.001

A	P(J|A)
T	0.90
F	0.05

A	P(M|A)
T	0.70
F	0.01

Cont…



- Prob. that both John and Mary call, the alarm sets off but no burglary nor earthquake:

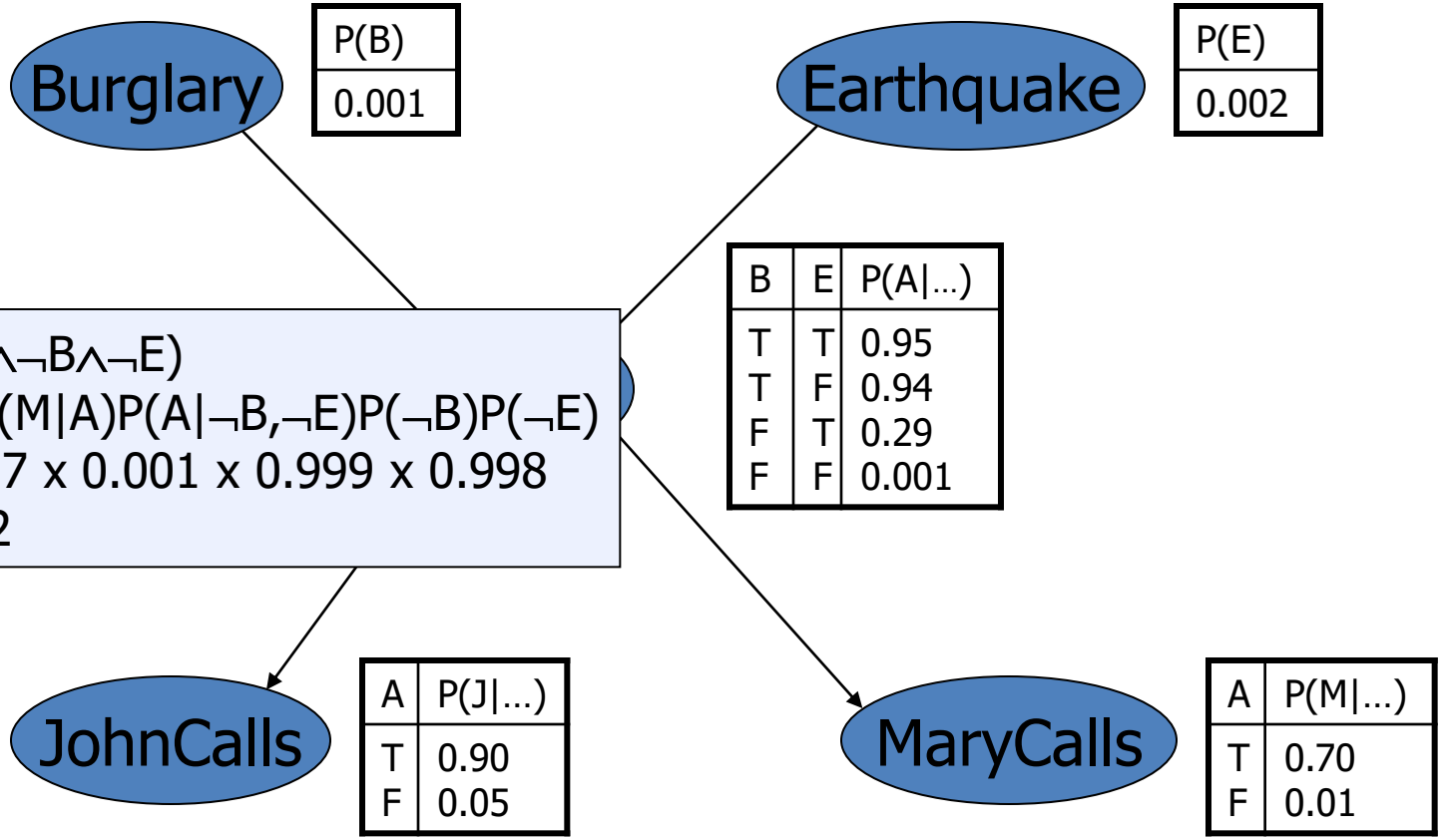
$=P(J \wedge M \wedge A \wedge \neg B \wedge \neg E)$

$= P(J|A)P(M|A)P(A|\neg B,\neg E)P(\neg B)P(\neg E)$

$= 0.9 \times 0.7 \times 0.001 \times 0.999 \times 0.998$

$= 0.00062$

Calculation of Joint Probability



P(B)
0.001

P(E)
0.002

B	E	P(A|...)
T	T	0.95
T	F	0.94
F	T	0.29
F	F	0.001

A	P(J|...)
T	0.90
F	0.05

A	P(M|...)
T	0.70
F	0.01

$P(J \wedge M \wedge A \wedge \neg B \wedge \neg E)$
$= P(J|A)P(M|A)P(A|\neg B,\neg E)P(\neg B)P(\neg E)$
$= 0.9 \times 0.7 \times 0.001 \times 0.999 \times 0.998$
$= 0.00062$

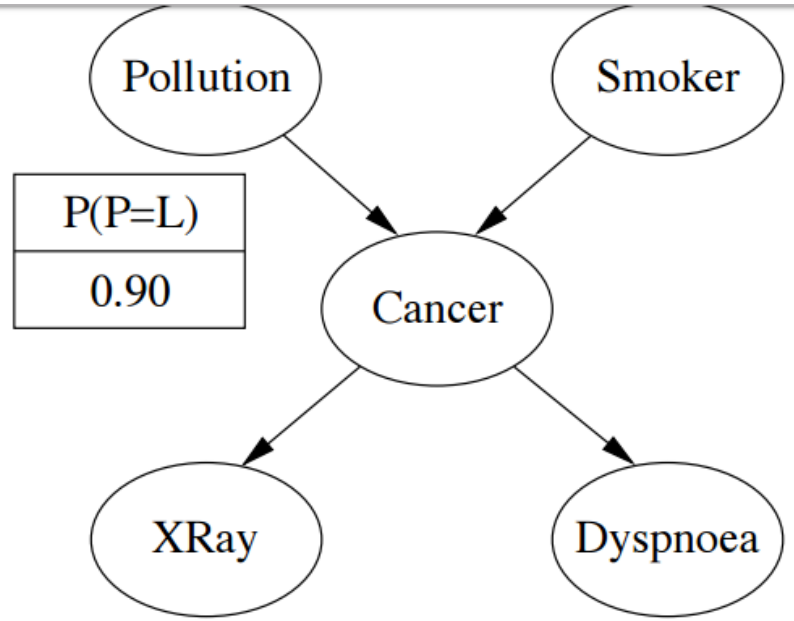
Practice

- Wet Pavement Problem- If the sprinkler is on, then the pavement is probably wet. If someone slips on pavement, it also provides evidence that the pavement is wet, that makes it more likely that the sprinkler is on or that it is raining.
- Life expectancy Problem-It is known that whether or not a person has cancer is directly influenced by whether she is exposed to second-hand smoke and whether (s)he smokes. Both of these things are affected by whether her parents smoke. Cancer reduces a person’s life expectancy.

Practice

- Lung Cancer Problem- Pollution and smoking are the factors those affect a patient's chance of having cancer. Similarly, having cancer affects the patient's breathing and the chances of having a positive X-ray result.
- Metastatic cancer: Metastatic cancer is a possible cause of brain tumours and is also an explanation for increased total serum calcium. In turn, either of these could explain a patient falling into a coma. Severe headache is also associated with brain tumours.

Lung Cancer Problem



P(S=T)
0.30

P(P=L)
0.90

P	S	P(C=T|P,S)
H	T	0.05
H	F	0.02
L	T	0.03
L	F	0.001

C	P(X=pos|C)
T	0.90
F	0.20

C	P(D=T|C)
T	0.65
F	0.30

FIGURE 2.1: A BN for the lung cancer problem.

Metastatic cancer

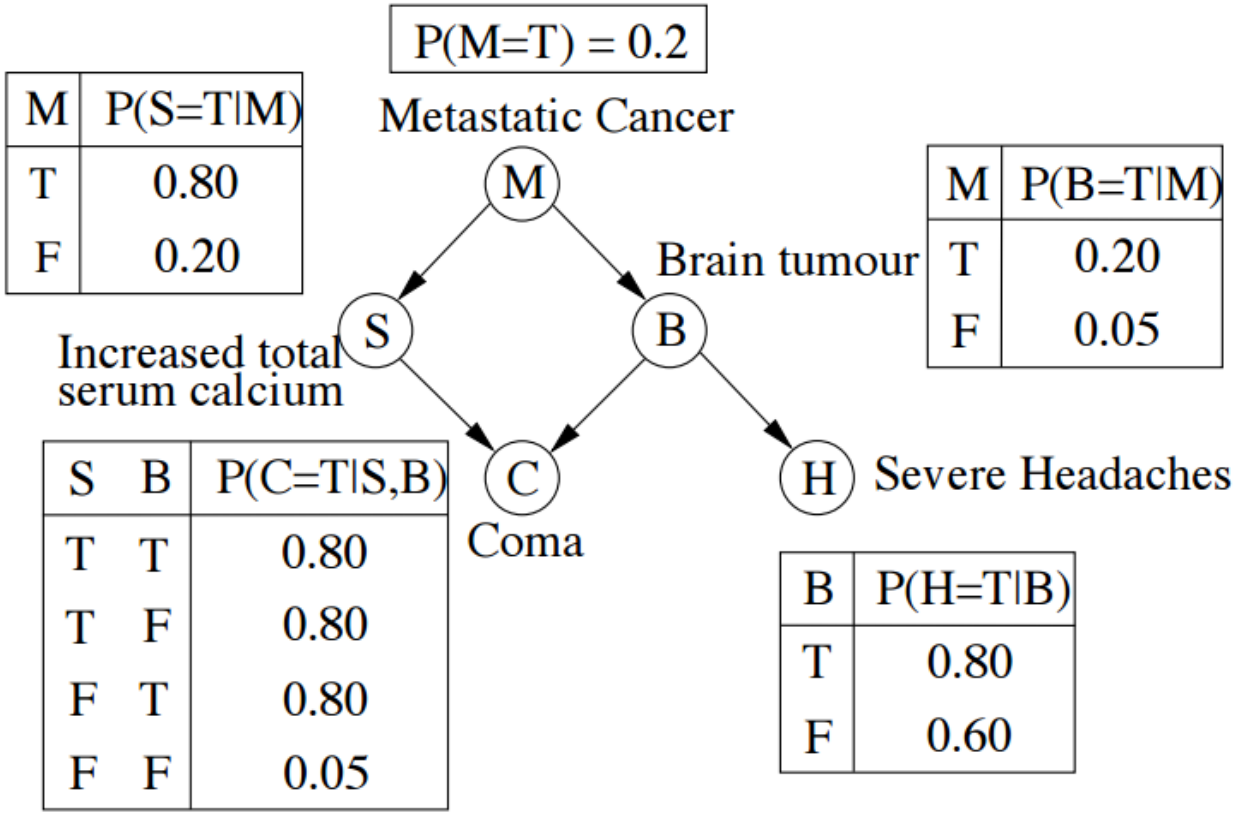


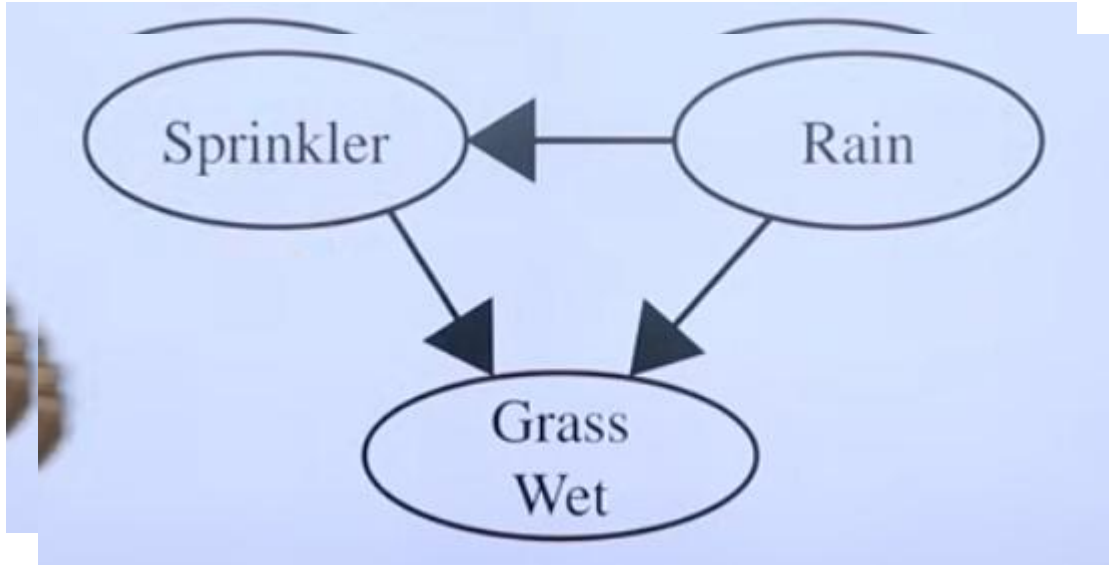
FIGURE 2.7: Metastatic cancer BN.

Summary

- Representing uncertainty is useful in knowledge bases.
- Probability provides a framework for managing uncertainty.
- Using a full joint distribution and probability rules, we can derive any probability relationship in a probability space.
- Number of required probabilities can be reduced through independence and conditional independence relationships
- Probabilities allow us to make better decisions by using decision theory and expected utilities.
- Rational agents cannot violate probability theory.

References

- Russel and Norving, “AIMA”, Second edition, PHI
- “Introducing Bayesian Networks”, http://www.csse.monash.edu.au/bai/book/BAI_Chapter2.pdf, last retrieved on April 16,2017


Sprinkler
Rain
Grass
Wet